

Zomato Restaurant Data Analysis

Documentation Report

Assignment: Exploratory Data Analysis using Python

Tools Used:	Python, Pandas, NumPy, Matplotlib, Seaborn
Dataset:	Zomato Restaurant Dataset (test_zomato.csv)
Records:	9,552 restaurants across 142 cities

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1. Dataset Overview

1.1 Description

The dataset contains information about restaurants listed on Zomato, including restaurant name, location, cuisines, average cost for two, ratings, pricing categories, and customer engagement metrics. The data spans restaurants across multiple cities and countries.

1.2 Size

The dataset contains **9,552 rows** (restaurants) and **22 columns**.

1.3 Column Details

Column Name	Type	Description
Restaurant ID	object	Unique identifier
Restaurant Name	object	Name of the restaurant
Country Code	float64	Numeric country code
City	object	City of the restaurant
Address	object	Full address
Locality	object	Locality within the city
Locality Verbose	object	Detailed locality
Longitude	object	Geographic longitude
Latitude	object	Geographic latitude
Cuisines	object	Cuisines served
Average Cost for two	object*	Estimated cost for two
Currency	object	Currency used
Has Table booking	object	Table booking available
Has Online delivery	object	Online delivery available
Is delivering now	object	Currently delivering
Switch to order menu	object	Order menu availability
Price range	float64	Price category (1-4)
Aggregate rating	float64	Customer rating (0-5)
Rating color	object	Color code for rating
Rating text	object	Rating category label
Votes	float64	Number of customer votes

* Originally string; converted to numeric for analysis.

2. Data Cleaning

2.1 Missing Values

Columns with missing (null) values:

Column	Missing
Locality	1
Locality Verbose	1
Longitude	1
Latitude	1
Cuisines	10
Average Cost for two	1
Currency	1
Has Table booking	1
Has Online delivery	1
Is delivering now	1
Switch to order menu	1
Price range	2
Aggregate rating	2
Rating color	2
Rating text	2
Votes	2

Cuisines has the most missing values (10). Most columns have only 1-2 missing entries.

2.2 Data Type Changes

The **Average Cost for two** column was converted from object (string) to numeric (float) using `pd.to_numeric(errors='coerce')` to enable calculations.

2.3 Duplicate Records

No duplicates found. All 9,552 entries are unique.

2.4 Data Quality Issues

The Has Online delivery column has one "Orange" entry (instead of Yes/No), and Has Table booking has one "3.2" entry. Both are data entry errors.

3. Data Analysis

3.1 Statistical Summary

Metric	Value
Total Restaurants	9,552
Unique Cities	142
Unique Cuisines	1,826
Average Aggregate Rating	2.67
Average Cost for Two	1,199.33
Min / Max Rating	0.0 / 4.9
Min / Max Votes	0 / 10,934
Average Votes	156.92
Average Price Range	1.80

3.2 Analysis Results

1. Average Rating: 2.67 / 5.0

2. Most Voted Restaurant: Toit — 10,934 votes

3. Average Cost for Two: 1,199.33

4. Ratings by Online Delivery:

Online Delivery	Avg Rating
Yes	3.25
No	2.47

Online delivery restaurants rate 0.78 points higher on average.

5. Ratings by Table Booking:

Table Booking	Avg Rating
Yes	3.44
No	2.56

Table booking restaurants rate 0.88 points higher on average.

6. Top 10 Cities:

Rank	City	Count
1	New Delhi	5,473
2	Gurgaon	1,118
3	Noida	1,080
4	Faridabad	251
5	Ghaziabad	25
6	Ahmedabad	21
7	Amritsar	21
8	Bhubaneswar	21
9	Guwahati	21
10	Lucknow	21

New Delhi holds 57% of all listings. Delhi-NCR region dominates.

4. Data Visualizations

4.1 Top 10 Cities — Restaurant Count (Bar Chart)

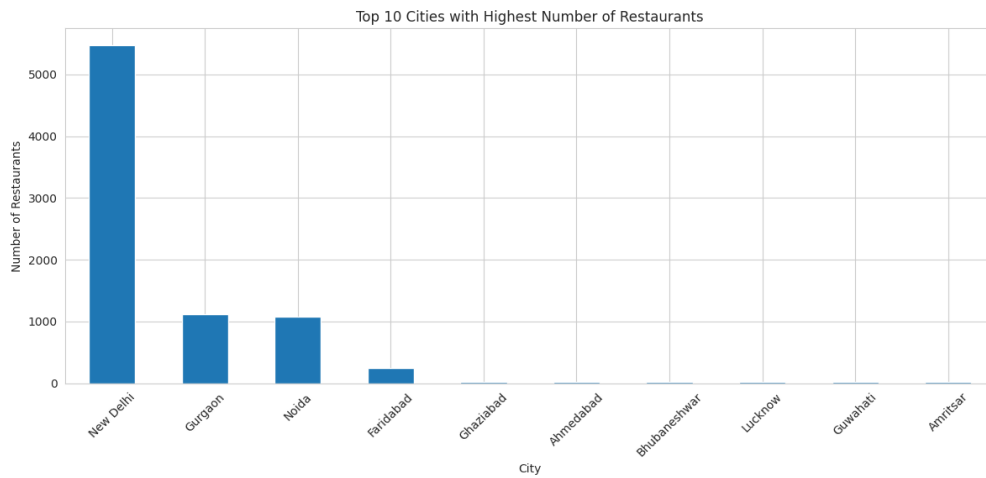


Figure 1: Top 10 Cities by Restaurant Count

New Delhi leads with 5,400+ restaurants. Gurgaon and Noida are the only others above 1,000. The remaining cities have very few listings — heavy Delhi-NCR concentration.

4.2 Distribution of Aggregate Ratings (Histogram)

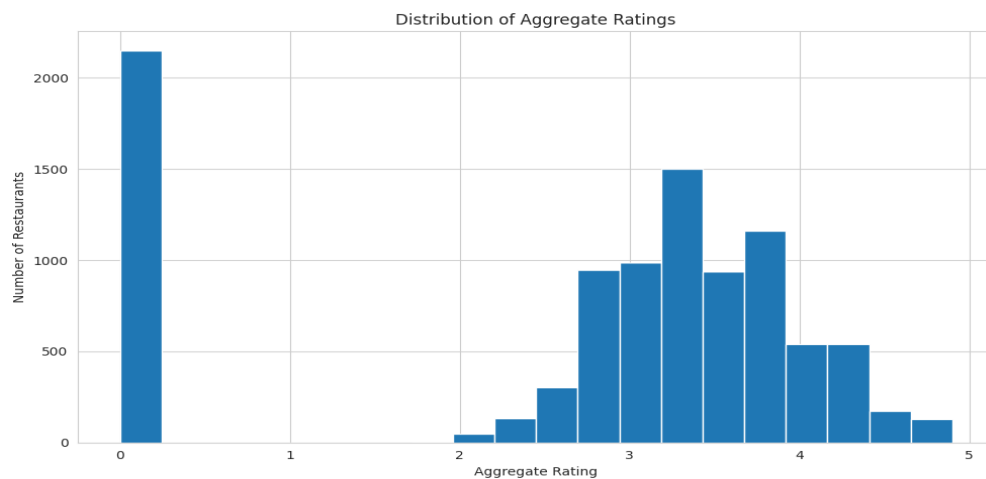


Figure 2: Distribution of Aggregate Ratings

Bimodal distribution: a spike near 0 (~2,150 unrated restaurants) and concentration in 3.0-4.0. Peak is around 3.2-3.4. Ratings above 4.5 or below 2.0 are rare.

4.3 Online Delivery (Bar Chart)

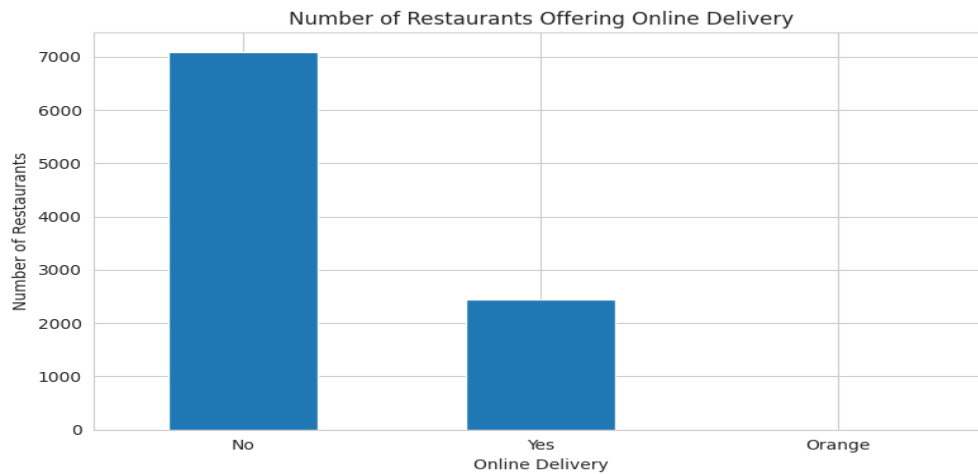


Figure 3: Online Delivery Availability

~7,100 restaurants lack online delivery; ~2,450 offer it. One "Orange" entry is a data error. Only ~25% provide delivery services.

4.4 Top 10 Most Common Cuisines (Bar Chart)

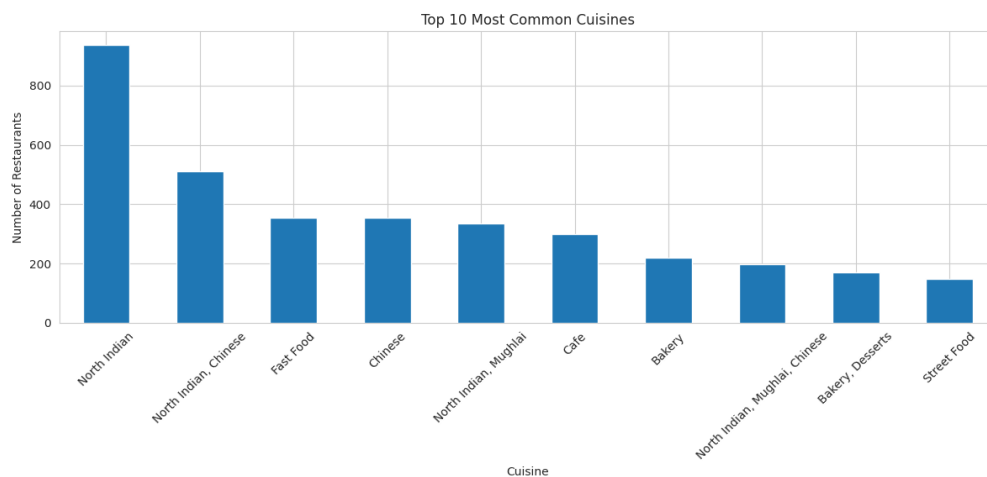


Figure 4: Top 10 Most Common Cuisines

North Indian leads (930+), then North Indian-Chinese (~510). Fast Food and Chinese each ~350. The pattern reflects the dataset's North India concentration.

4.5 Average Rating by Price Range (Line Chart)



Figure 5: Average Rating by Price Range

Clear positive correlation: budget restaurants (range 1) average 2.0; premium (range 4) average 3.82. The curve flattens between ranges 3-4, indicating diminishing returns.

5. Conclusion

5.1 Final Insights

- Highest presence:** New Delhi — 5,473 listings (57%).
- Most popular cuisine:** North Indian — 936 restaurants.
- Online Delivery effect:** Yes — avg 3.25 vs 2.47 without (+0.78).
- Table Booking effect:** Yes — avg 3.44 vs 2.56 without (+0.88).
- Key takeaway:** Dataset is Delhi-NCR heavy. ~2,150 unrated restaurants pull down averages. Higher price correlates with better ratings. Online delivery and table booking correlate positively with ratings — likely adopted by more established restaurants.

5.2 Business Observations

- Listings concentrate in Delhi-NCR, suggesting higher platform adoption there.
- North Indian cuisine dominance aligns with the geographic skew.
- Only ~25% offer delivery — significant growth opportunity for Zomato.
- Price-rating correlation: premium dining gets rated more favorably.
- Table booking and delivery restaurants rate higher — well-managed restaurants adopt these features.